

Soc Internship day 9 :Real incident case study (Brute Force Attack)

Scenario:

A system generated multiple alerts indicating failed login attempts on a server. After several attempts, a successful login was detected from the same IP

Log Data:

Apr 10 10:17:45 Failed password for user1 from 192.168.1.50
Apr 10 10:18:02 Failed password for user1 from 192.168.1.50
Apr 10 10:18:15 Failed password for user1 from 192.168.1.50
Apr 10 10:18:30 Failed password for user1 from 192.168.1.50
Apr 10 10:20:10 Accepted password for user1 from 192.168.1.50

1. Check the Timestamp

Failed logins happened between **10:17:45** and **10:18:30**.

2. Identify the User

The username is **user1**.

3. Identify the Source IP

The source IP is **192.168.1.10**.

4. Count Failed Attempts

There are **4 failed login attempts**.

5. Check for Successful Login

At **10:19:10**, login was successful.

6. Find the Pattern

Multiple failed attempts followed by one successful login.

7. Decide the Possible Attack

This may be a **brute-force attack** or a user entering the wrong password many times.

8. Take SOC Action

Block/check the IP, verify with the user, reset password if needed, and escalate to L2 if suspicious.

SOC Workflow Mapping

1. Alert Generation

Security tools detect suspicious activity and generate an alert.

Tools Involveds

- SIEM
- EDR/XDR
- Firewall
- IDS/IPS
- Antivirus

Example

Multiple failed login attempts detected from one IP address.

2. L1 Analyst – Triage

The L1 SOC analyst performs the first-level investigation.

Responsibilities

- Monitor alerts
- Validate alerts
- Identify false positives
- Check basic logs
- Assign severity level

Example Actions

- Check login logs
- Identify source IP
- Count failed attempts
- Verify affected user

Decision

- False Positive → Close alert
 - Suspicious Activity → Escalate to L2
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3. L2 Analyst – Investigation

The L2 analyst performs deeper technical analysis.

Responsibilities

- Detailed log analysis
- IOC investigation
- Threat hunting
- Malware analysis
- Scope identification

Example Actions

- Correlate firewall and endpoint logs
- Analyze attacker behavior
- Check lateral movement
- Identify affected systems

Decision

- Contained threat → Remediation
 - Advanced attack → Escalate to L3
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4. L3 Analyst – Advanced Response

The L3 analyst handles complex attacks and remediation.

Responsibilities

- Incident response
- Digital forensics
- Advanced malware analysis
- Threat containment
- Security improvements

Example Actions

- Isolate infected devices
 - Block malicious IPs/domains
 - Remove malware
 - Patch vulnerabilities
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5. Incident Response

The SOC team responds to the confirmed incident.

Actions

- Containment
- Eradication
- Recovery
- Monitoring

Example

- Reset passwords
 - Restore systems
 - Monitor for reinfection
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6. Documentation & Reporting

All findings and actions are documented.

Includes

- Timeline of attack
- Logs collected
- Impact assessment
- Actions taken
- Lessons learned

Student Questions:

1 What type of attack is this?

Brute-Force

2 Which IP is suspicious?

192.168.1.50

3 Was the attack successful?

Yes

4 What action should be taken?

Check if 192.168.1.50 belongs to a legitimate user/device.

- **Verify with user1** whether they were trying to log in.
- **Review login history** for unusual behavior.
- **Monitor post-login activity** (file access, privilege changes, downloads).
- **Check if brute-force behavior exists** (4 failed attempts then success).
- **Escalate to L2 analyst** if suspicious activity continues.
- **Document findings** in the SOC ticket.

Incident Analysis Report

Incident Title

Possible Brute Force Login Attempt

Incident Summary

A suspicious login pattern was identified during SOC monitoring. Multiple failed login attempts were observed from the same IP address, followed by one successful login to the same user account.

This behavior may indicate:

- User entering the wrong password multiple times
- Password guessing attempt
- Possible brute-force attack

Log Evidence

Apr 10 10:17:45 Failed password for user1 from 192.168.1.50
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Investigation Steps

1. Alert Identification

SOC monitoring detected multiple failed login attempts.

2. Log Review

The analyst reviewed authentication logs and observed:

- Username involved: **user1**
- Source IP: **192.168.1.10**
- Failed attempts: **4**
- Successful login: **1**

3. Pattern Analysis

The same IP address generated multiple failed logins and later authenticated successfully.

Pattern observed:

Failed Login → Failed Login → Failed Login → Failed Login → Successful Login

4. Threat Assessment

Possible scenarios:

Scenario 1 – Legitimate User

- User forgot password
- Multiple incorrect attempts
- Correct password entered later

Scenario 2 – Suspicious Activity

- Attacker attempted password guessing
- Credentials eventually obtained
- Possible brute-force compromise

5. Impact Analysis

Potential impact:

- Unauthorized account access
- Data exposure risk
- Privilege misuse possibility

Current impact level:

Medium Severity (Requires Validation)

6. SOC Actions Taken

L1 Analyst:

- Verified alert validity
- Reviewed authentication logs
- Identified suspicious login pattern

L2 Analyst:

- Correlated additional logs
- Reviewed account activity after login
- Checked for abnormal behavior

L3 Analyst (if escalated):

- Incident response support
- Containment actions
- Advanced forensic investigation

7. Recommended Actions

1. Verify login with user1
 2. Review account activity after login
 3. Check source IP reputation
 4. Monitor for additional failed attempts
 5. Reset password if compromise suspected
 6. Enable Multi-Factor Authentication (MFA)
 7. Document incident findings
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Final Conclusion

The event shows **4 failed password attempts followed by 1 accepted login from the same IP address**. This behavior could indicate a legitimate user mistake or a potential brute-force attack.

The SOC team should validate user activity and continue monitoring before closing the incident.

Prevention Methods

To reduce the risk of login attacks such as brute-force attempts, SOC teams can implement the following preventive measures:

1. Enable Multi-Factor Authentication (MFA)

- Require an additional verification step beyond passwords.
- Prevents attackers from accessing accounts even if passwords are compromised.

2. Implement Account Lockout Policy

- Temporarily lock accounts after multiple failed login attempts.
- Example: Lock account after **5 failed attempts for 15 minutes**.

3. Use Strong Password Policies

- Enforce complex passwords:
 - Minimum 12 characters
 - Uppercase and lowercase letters
 - Numbers
 - Special characters

4. Enable Login Monitoring

- Monitor authentication logs continuously.
- Configure alerts for unusual login behavior.

5. Apply Rate Limiting

- Restrict how many login attempts can occur within a short period.

6. Use IP Blocking or Firewall Rules

- Block suspicious IP addresses after repeated failed attempts.

7. Enable SIEM Correlation Rules

- Configure SIEM alerts for:
 - Multiple failed logins
 - Failed logins followed by successful login
 - Login attempts from unusual locations

8. User Awareness Training

- Train users to:
 - Use strong passwords
 - Avoid password reuse
 - Report suspicious account activity

9. Regular Security Audits

- Review authentication logs and access controls periodically.

10. Endpoint Protection

- Deploy EDR/XDR tools to detect suspicious activity on systems.

SOC Prevention Goal:

Detect Early → Prevent Access → Respond Quickly → Reduce Impact.

